Description

My notes about group theory. References below:

– 'Abstract algebra theory and applications, by T. Judson. https://github.com/twjudson/aata

– A book of abstract algebra, by C. Pinter: https://math.umd.edu/~jcohen/402/Pinter%20Algebra.pdf

Algebraic Structures

Question 1:

What is a relation R?

Answer:

A *relation* R is a subset of the cartesian product $A \times B$

$$R = \{(a, b) \in A \times B \mid aRb\}$$

Question 2:

What are operations?

Answer:

An operation is a

Question 3:

What are the group properties of a group G?

Answer:

Let U and V be in G. Then

1. *Inverse:* $U^{-1}, V^{-1} \in G$
2. *Closure:* $UV \in G$
3. *Identity:* $I \in G$

Question 4:

What is an abelian group?

Answer:

An *abelian group* (also called a *commutative group*) is a group G with binary operation R that also is endowed with the invariant of commutativity. That is, for $a, b \in G$ we have $a + b = b + a$

Question 5: What is a ring?

Answer:

A ring is a set A that is an abelian group with commutative binary operation R_1, that is also endowed under/with a second binary operation that satisfies the two invariant properties:

1. *Distributivity:* $aR_2(bR_1c) = (aR_2b)R_1(aR_2c)$
2. *Associativity:* $a(bc) = (ab)c$

It is the interaction between the two binary operations R_1 and R_2 that we get the full expressiveness and complexity of a ring (like polynomials for example).

Question 6: Examples of rings?

[Judson Pg 245]

Example:

Claim: the set of 2x2 matrices with entries of $\mathbb{R}$ is a ring under addition and multiplication.

Proof: Let $S = \{\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in \mathbb{R}\}$. Then we must show S is a ring under addition and multiplication. This means that S is an abelian group with operation addition and also endowed with the binary operation of multiplication that satisfies distributivity and associativity.

First we show S is an abelian group under addition. Observe that

1. Identity: 0 is the element for the addition operation; this is a trivial fact.
2. Associativity: Clearly for any $U, V, T \in S$ we have $U + (V + T) = (U + V) + T$
3. Inverse: Finally, it is trivially true that any $U \in S$ has an inverse $U^{-1} \in S$ since it is just the additive inverse of U.
4. Commutativity: For any $U, V \in S$, it is trivial to check that $U + V = V + U$

Therefore, indeed S is an abelian group under addition. Now we must show that S is closed under the multiplication operation and satisfies the invariant properties of distributivity and associativity. I leave such a proof to future me, for it is a tedious thing to type.

Question 7: What are the types of rings?

Ring with Unity/Identity: A ring S is said to be a ring with identity if there is an identity element in S, call it e_2, such that e_2 is an identity for the multiplication operation: $e_2 R_2 a = e_2 R_2 a = a$. Note I used the subscript 2 since e_1 is associated with the identity of the abelian group operation that S has (the addition operation)

Example

1. Set of binary strings under the $\vee$ and $\wedge$ operators

Commutative Ring: A ring is said to be a *commutative ring* if the R_2 binary operation is endowed with the commutativity invariant. That is, for $a, b \in S$ we have $aR_2b = bR_2a$

Examples of commutative rings:

1. Stabilizer group: $Stab(\psi)$. The stabilizer group is the set of stabilizer matrices and by definition they are commutative under multiplication, e.g. $ZI = IZ$

Question 8: What is a monoid?

Answer:

1. *Closure:* Operation R s.t. for $a, b \in M$ we have aRb
2. *Associativity*
3. *Identity*

Observe: Every group is a monoid, but not every monoid is a group. There is no inverse element in a monoid.

Example: String concat. Let Σ^* be a set of strings from alphabet Σ. Let R be the operation $aRb \iff a.append(b)$. Then (Σ, R) is a monoid since,

1. Closure: The concatenation of two strings is a string; therefore the operation has closure

2. Associativity: Let $a, b, c \in \Sigma^*$. We must show $(aRb)Rc = aR(bRc)$. In particular, observe $a.append(b) = ab$. So, $(ab).append(c)$ is abc. Moreover, $b.append(c)$ is ab and $a.append(bc)$ is abc. Therefore, the operation is associative.

3. Identity: Take the empty string $e = "" \in \Sigma^*$ to be the identity since clearly, for any $a \in \Sigma^*$ we have $a.append(e) = a.append(" ") = a$

We have shown $(\Sigma^*, concat)$ is closed under the operation, associative, and has an identity element. Therefore it is a monoid. □

Polynomials

First we define polynomials over a commutative ring A and then we define the set of all such polynomials. From those two definitions, we then show that the set of all polynomials in x is also a commutative ring with unity.

Question 9:

Define polynomial

Answer:

Let A be a commutative ring under addition and multiplication operations with unity. Then the object $p(x) = ax^1 + a_2x^2 + ... + a_na^n$ is called a *polynomial in x* over A.

Question 10: Define polynomial $A[x]$

Answer:

$A[x]$ is a set of polynomials in x whose coefficients are in the set A

Question 11: Prove the following

Claim: Let A be a commutative ring with unity. We claim $A[x]$ is a commutative ring.

Answer:

Proof:

Let A be a commutative ring with unity. We want to show that $A[X]$ is also a commutative ring with unity.

Part 1: Showing that $A[X]$ is an abelian group.

Let $p, q \in A[X]$, then they are of the form

$$p(x)$$

1. Identity: We want a polynomial $e_1 \in A[X]$ s.t. $p + e_1 = p$. Simply take e_1 as the monomial of degree 0 and coefficient zero. $e_1 \in A[X]$ by construction.

2. Associativity: $(q + p) + z = (q + p) + z$.

$p, q, z \in A[X]$ so we can write $p(x)$

3. Inverse
4. Commutativity

Maps

Question 12: What is a homomorphic map?

Answer:

Let G and H be groups with group operation $\cdot$ and $\circ$ respectively. Then a homomorphism is a map $L : G \to H$ such that $\forall x, y \in G$: $L(x \cdot y) = L(x) \circ L(y) \in Y$.

Question 13:

Example 1 (Judson Pg169): Let G be a group and $g \in G$. Define a map $L : Z \to G$ by $L(m) = g^m$ where $m \in \mathbf{Z}$. Show that L is a homomorphic map from group G to group Z

Answer:

A map is said to be homomorphic if for all $L(\alpha \cdot \alpha) = L(\alpha)L(\alpha)$. In this case, let $m, n \in Z$, then we have

$$L(m + n) = g^{m+n} = g^m g^n = L(m)L(n)$$

Hence, L is homomorphic.